

Ex 16 Experiment to understand EDA-Quantitative and Qualitative analysis.

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Name": ["Asha", "Ravi", "Kumar", "Priya", "John"],
    "Age": [20, 21, 19, 22, 20],
    "Marks": [85, 90, 78, 92, 88],
    "Gender": ["F", "M", "M", "F", "M"]
}

df = pd.DataFrame(data)

# Quantitative analysis
print("Quantitative Analysis:")
print(df[["Age", "Marks"]].describe())

# Qualitative analysis
print("\nQualitative Analysis:")
print(df["Gender"].value_counts())

# Visualization
df["Marks"].plot(kind="hist", bins=5)
plt.title("Marks Distribution")
plt.xlabel("Marks")
plt.show()
```

Quantitative Analysis:

```
...      Age      Marks
count    5.000000    5.000000
mean    20.400000   86.600000
std      1.140175    5.458938
min     19.000000   78.000000
25%     20.000000   85.000000
50%     20.000000   88.000000
75%     21.000000   90.000000
max     22.000000   92.000000
```

Qualitative Analysis:

Gender

M 3

F 2

Name: count, dtype: int64

